

Entropy-Governed Data Mortality in Complex Information Networks: A Reaction-Diffusion Framework with Entropy-Triggered Decay

1. Novelty Assessment and Literature Positioning

1.1 Gap Analysis

The existing literature addresses several adjacent ideas in isolation but **no unified framework** currently combines them:

Existing Work	What It Covers	What It Misses
Shannon entropy / information theory	Entropy as a measure of uncertainty	No lifecycle or temporal dynamics for data objects
Epidemic models on networks (SIR/SIS)	Spreading dynamics with recovery	No entropy-based feedback; “recovery” is not data deletion governed by exposure
Reaction-diffusion on graphs	Continuous diffusion + local reaction terms	No information-theoretic mortality condition
Data lifecycle management (DLM)	Policy-based retention/deletion	No mathematical model; rule-based, not dynamical
ERASE / ERASERS (IEEE)	Entropy of individual blocks for secure deletion	Single-node, no network context; entropy of content, not distribution
Differential privacy	Noise injection to bound information leakage	Operates at query time, not on data persistence itself
Anti-entropy protocols	Consistency repair via gossip	“Entropy” refers to inconsistency, not information-theoretic exposure
Time-decaying information diffusion (NIH, 2020s)	Exponential decay factors in social contagion	Ad hoc decay; not governed by a computed entropy threshold

The novelty of EGDM lies in coupling three elements that have not previously been unified:

1. **Graph-Laplacian diffusion** of data across a network,

2. **Real-time computation of the distributional entropy** of that data,
3. **An entropy-triggered mortality mechanism** that feeds back into the diffusion dynamics.

This creates a **closed-loop dynamical system** where the macroscopic statistical property (entropy) governs microscopic state changes (node-level data deletion) — an approach with no direct precedent.

1.2 Research Value

- **Theoretical:** Introduces a new class of reaction-diffusion systems where the reaction term is controlled by a global entropy functional — bridging statistical physics, information theory, and network science.
- **Applied:** Provides a principled, tunable mechanism for automatic data expiration in distributed systems, with direct relevance to GDPR-style data retention, AI training data governance, and privacy-preserving architectures.
- **Methodological:** The framework is amenable to both analytical treatment (stability, bifurcation, equilibrium) and computational simulation, making it suitable for an applied mathematics audience.

2. Refined Research Title

Entropy-Governed Data Mortality in Complex Information Networks: A Reaction-Diffusion Framework with Entropy-Triggered Decay

Alternative shorter title for conferences:

Data Mortality via Entropy Thresholds on Complex Networks

3. Abstract (245 words)

Modern digital infrastructures are designed around indefinite data persistence, creating systemic risks including privacy erosion, surveillance accumulation, and storage overload. We propose Entropy-Governed Data Mortality (EGDM), a mathematical framework in which the lifetime of distributed data is dynamically controlled by the entropy of its spatial distribution across a complex network. The digital infrastructure is modelled as a weighted graph $G = (V, E, W)$, and data propagation follows a graph-Laplacian diffusion process. At each instant, the Shannon entropy of the normalised data distribution is computed as a macroscopic exposure metric. When this entropy exceeds a critical threshold H_c , an entropy-triggered decay term is activated, reducing data concentrations at over-exposed nodes according to a nonlinear feedback law.

The resulting system is a novel class of reaction-diffusion equation on graphs in which the reaction term is governed by a global entropy functional rather than local state variables alone. We establish conditions for the existence and uniqueness of solutions, characterise the equilibrium states of the system, and prove that the critical entropy threshold induces a transcritical bifurcation separating a data-persistent regime from a data-mortal regime. We derive explicit relationships between H_c , the spectral gap of the graph Laplacian, and the asymptotic data distribution. Numerical simulations on Erdős–Rényi, Barabási–Albert, and empirical network topologies confirm the analytical predictions and demonstrate that EGDM produces controlled, graceful data decay rather than abrupt deletion. The framework offers a mathematically grounded alternative to policy-based data retention, with applications in privacy engineering, distributed storage, and AI data governance.

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5. Mathematical Formalisation

5.1 Network Representation

Let $G = (V, E, W)$ be a connected, undirected, weighted graph with $n = |V|$ nodes. The weighted adjacency matrix is $W = [w_{ij}]$ where $w_{ij} > 0$ if $(i, j) \in E$ and $w_{ij} = 0$ otherwise. The degree of node i is $d_i = \sum_j w_{ij}$. The graph Laplacian is

$$L = D - W, \quad D = \text{diag}(d_1, \dots, d_n).$$

5.2 State Vector

Let $\mathbf{x}(t) = (x_1(t), \dots, x_n(t))^T \in \mathbb{R}_{\geq 0}^n$ represent the **data concentration** at each node at time t . We require $x_i(t) \geq 0$ for all i, t .

Define the **total data mass**:

$$M(t) = \sum_{i=1}^n x_i(t).$$

When $M(t) > 0$, define the **normalised distribution**:

$$p_i(t) = \frac{x_i(t)}{M(t)}.$$

5.3 Pure Diffusion Dynamics

Without mortality, data propagates via the graph Laplacian:

$$\frac{d\mathbf{x}}{dt} = -\alpha L\mathbf{x}, \quad \alpha > 0,$$

which expands component-wise to

$$\frac{dx_i}{dt} = \alpha \sum_j w_{ij}(x_j - x_i).$$

This is a continuous-time consensus/heat equation on the graph. Under this dynamics alone, $M(t)$ is conserved and $\mathbf{x}(t) \rightarrow \frac{M(0)}{n}\mathbf{1}$ as $t \rightarrow \infty$ (uniform distribution), with convergence rate governed by the **algebraic connectivity** $\lambda_2(L)$ (Fiedler value).

5.4 Entropy Exposure Metric

The Shannon entropy of the normalised distribution is

$$H(\mathbf{p}(t)) = -\sum_{i=1}^n p_i(t) \ln p_i(t),$$

with the convention $0 \ln 0 = 0$.

Properties: - $H \in [0, \ln n]$. - $H = 0$ iff data is concentrated at a single node. - $H = \ln n$ iff data is uniformly distributed (maximum exposure). - Under pure diffusion, $H(t)$ is **monotonically non-decreasing** (we prove this below).

5.5 Entropy-Triggered Mortality Mechanism

Define the **mortality activation function**:

$$\sigma(H) = \left[\frac{H - H_c}{H_{\max} - H_c} \right]_+^\beta$$

where: - $H_c \in (0, \ln n)$ is the **critical entropy threshold**, - $H_{\max} = \ln n$ is the maximum possible entropy, - $[\cdot]_+ = \max(\cdot, 0)$ denotes the positive part, - $\beta \geq 1$ controls the **sharpness** of mortality onset ($\beta = 1$: linear; $\beta \gg 1$: switch-like).

The mortality term removes data from nodes in proportion to their current concentration:

$$R_i(\mathbf{x}, H) = -\mu \sigma(H) x_i, \quad \mu > 0,$$

where μ is the **mortality rate constant**.

5.6 Full EGDM Dynamical System

Combining diffusion, mortality, and an optional exogenous data injection source $s_i(t) \geq 0$:

$$\boxed{\frac{dx_i}{dt} = \underbrace{\alpha \sum_j w_{ij}(x_j - x_i)}_{\text{diffusion}} \underbrace{- \mu \sigma(H(\mathbf{p})) x_i}_{\text{entropy-triggered mortality}} + \underbrace{s_i(t)}_{\text{injection}}}$$

In matrix form:

$$\frac{d\mathbf{x}}{dt} = -\alpha L\mathbf{x} - \mu \sigma(H(\mathbf{p})) \mathbf{x} + \mathbf{s}(t).$$

Key structural feature: The mortality term depends on $H(\mathbf{p})$, which is a *global* functional of the entire state vector. This makes the system **non-local** — each node's dynamics depends on the distribution across all nodes — distinguishing EGDM from standard reaction-diffusion systems where reactions are purely local.

5.7 Evolution of Total Mass

Summing over all nodes:

$$\frac{dM}{dt} = -\mu \sigma(H) M(t) + S(t), \quad S(t) = \sum_i s_i(t).$$

Since $\mathbf{1}^T L = \mathbf{0}$, diffusion conserves mass. Mortality reduces mass at rate $\mu \sigma(H)$. In the source-free case ($S = 0$):

$$M(t) = M(0) \exp\left(-\mu \int_0^t \sigma(H(\tau)) d\tau\right).$$

5.8 Entropy Dynamics

Differentiating $H(\mathbf{p})$ along trajectories of the system yields (after calculation):

$$\frac{dH}{dt} = - \underbrace{\frac{\alpha}{M} \sum_{(i,j) \in E} w_{ij}(x_j - x_i)(\ln p_j - \ln p_i)}_{\text{diffusion contribution } \Delta_D \geq 0} + \underbrace{(\text{source and mortality correction terms})}_{\text{depend on } \mathbf{s} \text{ and } \sigma}$$

The diffusion contribution $\Delta_D \geq 0$ because $(x_j - x_i)$ and $(\ln p_j - \ln p_i)$ always have the same sign (both are monotone in p). This proves entropy is non-decreasing under pure diffusion.

5.9 Stochastic Extension

For modelling noisy environments, replace the deterministic ODE with an Itô SDE:

$$dx_i = \left[\alpha \sum_j w_{ij}(x_j - x_i) - \mu\sigma(H)x_i + s_i \right] dt + \eta\sqrt{x_i} dB_i(t)$$

where $\{B_i(t)\}$ are independent Wiener processes and $\eta > 0$ scales the noise. The $\sqrt{x_i}$ factor ensures non-negativity (analogous to Cox–Ingersoll–Ross processes).

6. Potential Theorem Statements

Theorem 1 (Monotonicity of Entropy under Pure Diffusion)

Let $\mathbf{x}(t)$ evolve under $\dot{\mathbf{x}} = -\alpha L\mathbf{x}$ with $\mathbf{x}(0) \in \mathbb{R}_{>0}^n$ and $\alpha > 0$. Then $H(\mathbf{p}(t))$ is monotonically non-decreasing, and $dH/dt = 0$ if and only if $\mathbf{p}$ is the uniform distribution.

Proof sketch. Use the log-sum inequality and the fact that L is positive semi-definite with kernel $\text{span}(\mathbf{1})$. The entropy production $\Delta_D = -\frac{\alpha}{M}\mathbf{x}^T L(\ln \mathbf{p})$ can be shown to equal $\frac{\alpha}{M} \sum_{(i,j)} w_{ij}(p_i - p_j)(\ln p_i - \ln p_j) \geq 0$ by the elementary inequality $(a - b)(\ln a - \ln b) \geq 0$.

Theorem 2 (Mass Extinction in the Source-Free Case)

Consider the EGDM system with $\mathbf{s} \equiv 0$ and $\beta = 1$. If $H(\mathbf{p}(0)) > H_c$, then $M(t) \rightarrow 0$ as $t \rightarrow \infty$.

Proof sketch. By Theorem 1, $H(t) \geq H(0) > H_c$ for all $t \geq 0$ (since the mortality term only removes data proportionally and cannot decrease entropy

below its initial value when acting uniformly). Therefore $\sigma(H(t)) \geq \sigma(H(0)) > 0$ for all t . From the mass equation, $dM/dt \leq -\mu\sigma(H(0))M$, giving exponential decay $M(t) \leq M(0)e^{-\mu\sigma(H(0))t} \rightarrow 0$.

Theorem 3 (Existence of Non-Trivial Equilibrium with Constant Source)

Let $\mathbf{s} = \mathbf{s}_0$ be constant with $S_0 = \sum_i s_{0,i} > 0$. Then the EGDM system admits at least one equilibrium $\mathbf{x}^* > 0$ satisfying $-\alpha L\mathbf{x}^* - \mu\sigma(H(\mathbf{p}^*))\mathbf{x}^* + \mathbf{s}_0 = 0$.

Proof sketch. Define $F(\mathbf{x}) = -\alpha L\mathbf{x} - \mu\sigma(H(\mathbf{p}))\mathbf{x} + \mathbf{s}_0$. On the compact convex set $\Omega_R = \{\mathbf{x} \in \mathbb{R}_{\geq 0}^n : M \leq R\}$ for sufficiently large R , one shows F points inward on $\partial\Omega_R$ (since $dM/dt < 0$ for large M) and upward on faces where $x_i = 0$ (since $s_{0,i} > 0$). By Brouwer's fixed-point theorem, a fixed point exists in the interior.

Theorem 4 (Transcritical Bifurcation at H_c)

Consider the EGDM system with constant uniform source $s_i = s_0/n$. As H_c is varied, the system undergoes a transcritical bifurcation: for $H_c > H^*$ (where H^* is the entropy of the source-only equilibrium), the mortality term is inactive and the equilibrium has $M^* = S_0/0 \rightarrow \infty$ governed by diffusion-source balance; for $H_c < H^*$, a new stable equilibrium emerges with $M^* < M_{no-mort}^*$, and the two branches exchange stability at $H_c = H^*$.

Theorem 5 (Spectral Gap Controls Mortality Activation Time)

Under pure diffusion from a localised initial condition $x_i(0) = M_0\delta_{i,i_0}$, the time t^* at which $H(\mathbf{p}(t^*)) = H_c$ satisfies

$$t^* = \Theta\left(\frac{\ln n - H_c}{\alpha\lambda_2(L)}\right)$$

where $\lambda_2(L)$ is the algebraic connectivity (Fiedler value) of G .

Proof sketch. The rate of entropy increase near the initial condition is dominated by the second eigenmode of L , giving $H(t) \approx \ln n - Ce^{-2\alpha\lambda_2 t}$. Solving $H(t^*) = H_c$ yields the scaling.

7. Connections to Established Mathematical Frameworks

7.1 Reaction-Diffusion Systems on Graphs

The EGDM system has the general form $\dot{\mathbf{x}} = -\alpha L\mathbf{x} + \mathbf{R}(\mathbf{x})$, placing it within the class of reaction-diffusion systems on discrete domains. The novelty is that $\mathbf{R}$ depends on a **global entropy functional** rather than local concentrations.

Related frameworks: - Fisher–KPP equation on graphs (logistic local reaction) - Turing pattern formation on networks - Allen–Cahn and Cahn–Hilliard equations on graphs

7.2 Epidemic Models (SIR/SIS)

EGDM shares structural similarities with compartmental epidemic models:

Epidemic analogy	EGDM counterpart
Susceptible $\rightarrow$ Infected	Node without data $\rightarrow$ node with data (diffusion)
Infected $\rightarrow$ Removed	Node with data $\rightarrow$ data deleted (mortality)
Basic reproduction number R_0	Ratio $\alpha\lambda_{\max}/(\mu\sigma)$

However, in EGDM the “removal rate” is **not constant** but depends on the global entropy — this is analogous to an SIR model with **awareness-dependent recovery**, a connection to behavioural epidemiology.

7.3 Entropy Production in Non-Equilibrium Statistical Physics

The entropy dynamics of EGDM parallel the second law of thermodynamics for Markov chains. Under pure diffusion, H increases (entropy production ≥ 0), mimicking irreversible relaxation to equilibrium. The mortality term acts as an **external intervention** that modifies the entropy production, analogous to Maxwell’s demon or feedback control in thermodynamic systems.

7.4 Lyapunov Functions and Gradient Flows

The pure-diffusion dynamics $\dot{\mathbf{x}} = -\alpha L\mathbf{x}$ can be written as a gradient flow with the energy $E(\mathbf{x}) = \frac{\alpha}{2}\mathbf{x}^T L\mathbf{x}$. The full EGDM system may admit a modified Lyapunov function of the form

$$V(\mathbf{x}) = \frac{\alpha}{2}\mathbf{x}^T L\mathbf{x} + \mu\Phi(H(\mathbf{p}), M)$$

where Φ encodes the entropy–mortality interaction. Identifying such a function would guarantee global stability.

7.5 Mean-Field Games and Optimal Control

The entropy threshold H_c can be reinterpreted as a **constraint in an optimal control problem**: minimise a cost functional (e.g., storage cost + privacy risk) subject to network diffusion dynamics. The mortality rate $\mu\sigma(H)$ then emerges as the optimal feedback control law. This opens connections to: - Pontryagin's maximum principle on networks - Hamilton–Jacobi–Bellman equations for distributed systems - Mean-field games if nodes are strategic agents

8. Suggested Simulation Experiments

Experiment 1: Entropy Trajectory under Pure Diffusion

- **Setup:** Generate $ER(n, p)$ and $BA(n, m)$ graphs for $n \in \{50, 200, 1000\}$. Initialise data at a single node. Run pure diffusion ($\mu = 0$).
- **Measure:** $H(t)$ trajectory, convergence time to $H = \ln n - \varepsilon$.
- **Purpose:** Validate Theorem 1 and Theorem 5; verify spectral gap scaling.

Experiment 2: Mortality Activation and Mass Decay

- **Setup:** Same graphs, now with mortality ($\mu > 0$, $H_c = 0.7 \ln n$). No sources.
- **Measure:** Time to mortality activation t^* , mass decay trajectory $M(t)$, node-level deletion patterns.
- **Purpose:** Validate Theorem 2; compare analytical t^* prediction with simulation.

Experiment 3: Equilibrium with Constant Source

- **Setup:** Uniform constant source $s_i = s_0/n$. Vary H_c across $(0, \ln n)$.
- **Measure:** Equilibrium mass M^* , equilibrium entropy H^* , bifurcation diagram $M^*(H_c)$.
- **Purpose:** Validate Theorem 3 and Theorem 4; visualise transcritical bifurcation.

Experiment 4: Sensitivity to Network Topology

- **Setup:** Fix α, μ, H_c, s_0 . Compare across: complete graph, star graph, ring lattice, ER, BA, Watts–Strogatz small-world, empirical Internet AS topology.
- **Measure:** t^* , M^* , entropy trajectory shape, spatial heterogeneity of deletion.
- **Purpose:** Characterise how topology modulates mortality dynamics.

Experiment 5: Parameter Sensitivity Analysis

- **Setup:** Fix an $ER(100, 0.1)$ graph with constant source. Vary each parameter independently: $\alpha \in [0.01, 10]$, $\mu \in [0.01, 10]$, $H_c \in [0.2 \ln n, 0.95 \ln n]$, $\beta \in \{1, 2, 5, 20\}$.
- **Measure:** Heat maps of M^* and H^* as functions of parameter pairs.
- **Purpose:** Identify practically useful parameter regimes.

Experiment 6: Stochastic Simulations

- **Setup:** Run the SDE version (Section 5.9) with $\eta \in \{0.01, 0.1, 0.5\}$. Euler–Maruyama integration. 500 sample paths per configuration.
- **Measure:** Distribution of t^* , variance of M^* , probability of premature mortality activation.
- **Purpose:** Assess robustness of deterministic predictions under noise.

Experiment 7: Comparison with Baseline Deletion Strategies

- **Setup:** Compare EGDM against: (a) fixed-TTL deletion (data deleted after constant time), (b) random deletion (each node deletes with constant probability), (c) no deletion.
- **Measure:** Time-averaged entropy, total mass, “privacy score” (peak entropy reached), deletion efficiency (data removed per unit privacy risk).
- **Purpose:** Demonstrate the advantage of entropy-governed mortality over naïve strategies.

9. Possible Datasets and Synthetic Simulations

Synthetic Networks

Model	Parameters	Rationale
Erdős–Rényi $G(n, p)$	$n = 100\text{--}1000$, $p = 2 \ln n / n$	Baseline random graph; well-understood spectral properties
Barabási–Albert	$n = 100\text{--}1000$, $m = 3$	Scale-free topology mimicking real infrastructure
Watts–Strogatz	$n = 200$, $k = 6$, $p = 0.1$	Small-world property common in social/communication networks
Random geometric graph	$n = 200$, $r = \sqrt{\ln n / n}$	Spatial networks mimicking sensor/IoT deployments

Model	Parameters	Rationale
Stochastic block model	4 communities, mixing parameter μ	Community structure in organisational data silos

Empirical Topologies

Dataset	Source	Nodes	Why
CAIDA AS-relationships	CAIDA	~70k	Real Internet backbone; natural data
Stanford SNAP email-Eu-core	SNAP	1,005	transit network Organisational communication patterns
Facebook ego-networks	SNAP	4,039	Social network with community structure
Airport network (OpenFlights)	OpenFlights.org	~3,400	Transportation analogy for data routing

10. Recommended Journals and Conferences

Q1 Journals (Primary Targets)

Journal	Publisher	Focus	Impact Factor
<i>Chaos: An Interdisciplinary Journal of Nonlinear Science</i>	AIP	Complex systems, nonlinear dynamics	~2.9
<i>Journal of Complex Networks</i>	Oxford	Network science, graph dynamics	~2.1
<i>Physical Review E</i>	APS	Statistical physics, complex systems	~2.4

Journal	Publisher	Focus	Impact Factor
<i>SIAM Journal on Applied Dynamical Systems</i>	SIAM	Dynamical systems, applied math	~2.1
<i>IEEE Transactions on Information Theory</i>	IEEE	Information theory fundamentals	~2.5
<i>Chaos, Solitons & Fractals</i>	Elsevier	Nonlinear science, complexity	~5.3
<i>Journal of Statistical Mechanics: Theory and Experiment</i>	IOP	Statistical mechanics on networks	~2.2

Q2 Journals (Secondary Targets)

Journal	Publisher	Focus
<i>Applied Mathematics and Computation</i>	Elsevier	Computational applied math
<i>Physica A: Statistical Mechanics and its Applications</i>	Elsevier	Statistical physics, networks
<i>Entropy</i> (MDPI)	MDPI	Information theory, complexity (open access)
<i>Journal of Mathematical Biology</i>	Springer	If biological/ecological analogy is emphasised
<i>Journal of Computational Science</i>	Elsevier	Computational modeling

Conferences

Conference	Community	Format
NetSci	Network Science Society	Contributed/invited talks
International Conference on Complex Networks (CompleNet)	Complex networks	Full papers

Conference	Community	Format
IEEE Symposium on Security and Privacy (S&P)	Cybersecurity (if applied angle is strong)	Full papers
Conference on Complex Systems (CCS)	Complex Systems Society	Contributed talks
SIAM Conference on Applications of Dynamical Systems	SIAM	Minisymposia

11. Summary of Suggested Improvements over Original Formulation

Original Formulation	Improvement
Adjacency-based diffusion $\sum A_{ij}(x_j - x_i)$	Formalised using graph Laplacian $-\alpha L\mathbf{x}$ with weighted edges
Binary mortality: delete when $H > H_c$	Smooth activation $\sigma(H) = [(H - H_c)/(H_{\max} - H_c)]_+^\beta$ with tuneable sharpness
No mass conservation analysis	Explicit mass ODE; exponential decay in source-free case
No entropy dynamics	Derived dH/dt along system trajectories; proved monotonicity under diffusion
No equilibrium analysis	Fixed-point analysis with Brouwer existence proof; bifurcation at H_c
No spectral connection	Spectral gap λ_2 governs mortality activation time
Deterministic only	Itô SDE extension with CIR-type noise preserving non-negativity
No connection to epidemics	SIR analogy with entropy-dependent recovery rate
No relation to control theory	Reinterpretation as entropy-constrained optimal control problem
No baseline comparison	Seven simulation experiments including comparison with TTL and random deletion

12. Notation Summary

Symbol	Meaning
$G = (V, E, W)$	Weighted undirected graph (network)
$n = \ V\ $	Number of nodes
$W = [w_{ij}]$	Weighted adjacency matrix
$L = D - W$	Graph Laplacian
$\lambda_2(L)$	Algebraic connectivity (Fiedler value)
$\mathbf{x}(t) \in \mathbb{R}_{\geq 0}^n$	Data concentration vector
$M(t) = \sum_i x_i(t)$	Total data mass
$p_i(t) = x_i(t)/M(t)$	Normalised data distribution
$H(\mathbf{p})$	Shannon entropy of distribution
H_c	Critical entropy threshold
$\sigma(H)$	Mortality activation function
α	Diffusion rate
μ	Mortality rate constant
β	Mortality sharpness exponent
$\mathbf{s}(t)$	Exogenous data injection rate
η	Stochastic noise intensity

Document prepared as a publication-quality research framework for the EGDM project.